FEDERAL SPECIFICATION

MELAMINE MOLDED PLASTICS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers melamine molded products. The chief characteristics of melamine molded products are their thermosetting nature, rigidity, abrasion resistance, water and chemical resistance, nonflammability, heat resistance up to 210°-250° F., arc resistance, and color stability. Melamine molded products do not impart taste or odor to foods. Impact strength is dependent upon the type of reinforcing filler used with the resins.

1.2 Classification.

1.2.1 *Types.*—Melamine plastics covered by this specification shall be of the types listed in table I, as specified in the contract or purchase order:

2. APPLICABLE SPECIFICATIONS

2.1 The following Federal Specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

L–P–406—Plastics, Organic: General Specifications, Test Methods.

2.2 Specifications and other publications applicable only to individual departments are listed in section 7.

TABLE I.—*Types of melamine molded plastics*

Type	Filler
I. General-purpose material, suitable for general, mechanical and electrical applications, and tableware. Available in colors, white, and black.	Alpha-cellulose
II. Heat-resistant and electrical material.	Mineral
III. Impact resistant material.	Chopped cotton cloth
IV. Melamine-phenol material, having impact strength greater than type III.	Chopped cotton cloth
V. High translucency material, particularly suited for small pearlescent moldings such as buttons.	None
VI. Industrial use material, suitable for general, mechanical and electrical applications, particularly at elevated temperatures.	Cellulose
VII. Industrial use material, having improved dimensional stability and insert properties.	Alpha-cellulose

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3. REQUIREMENTS

3.1 Material.—The melamine molding compounds used to mold the molded parts shall be composed of a melamine-formaldehyde resin binder, with or without small amounts of other resins, that has been intimately combined in the uncured or partially cured condition with fillers, pigments, and other chemical agents.

3.2 Design and dimensions.—The design and dimensions of the molded part shall be as specified in the contract or purchase order. All molded parts shall be within the dimensional tolerances specified in the contract or shown on the drawings. Accuracy of molding shall be such as to assure interchangeability of all parts made to the same drawings.

3.3 Inserts.—All inserts, holes, lugs, etc., shall be correctly spaced as specified in the contract or shown on applicable drawings. Inserts shall not be battered or damaged in any way, shall be free from defects of material or workmanship, and shall not be coated with molded material in such a manner as to lead to imperfect electrical contact or poor mechanical fit.

3.4 Threads.—All threaded sections, whether metal inserts or threads tapped or molded into the molded part, shall have smooth clean threads free from nicks, tears, or other damage.

3.5 Color and opacity.—The color and opacity of the molded part shall be as specified in the contract or purchase order.

3.6 Finish.—The molded part shall have a finish as specified in the contract or purchase order. When a smooth surface is specified, it shall be similar to a polished finish. Unless otherwise specified in the contract or purchase order, no surface shall be painted and no machining of surfaces shall be permitted except that flash, burrs, or projecting ridges

TABLE II.—*Physical requirements for compression molded test specimens of melamine plastics*

	Type						
	I	II	III	IV	V	VI	VII
Specific gravity,[1] 23°/23° C. (73.5°/73.5° F.):							
Minimum	1.47	1.76	1.47	1.47	1.46	1.42	1.40
Maximum	1.53	1.80	1.53	1.53	1.50	1.48	1.46
Flexural strength,[2] minimum, p.s.i.	12,000	8,000	12,000	11,000	11,000	8,000	8,000
Flexural deflection,[2] minimum, inch	0.110	0.055	0.100	0.100	0.110	0.095	0.075
Impact strength, Izod,[2] minimum, ft.-lb. per inch of notch	0.25	0.30	0.55	1.0	.....	0.25	0.30
Water absorption,[3] (24 hr., immersion), maximum weight gain, percent	0.4	0.15	0.6	0.5	0.35	0.60	0.60
Dielectric strength,[1] minimum, volts per mil:							
Short-time test	300	350	250	130	.....	350	350
Step-by-step test	250	250	100	.....	.....	200	200
Loss factor,[2] maximum:							
60 cycles per sec.	0.78	1.0	0.97	.....	.....	0.22	0.22
1,000 cycles per sec.	0.31	0.67	0.39	.....	.....	0.26	0.27
1,000,000 cycles per sec.	0.23	0.31	0.24	.....	.....	0.19	0.28
Arc resistance,[1] minimum, sec.	120	120	125	4	125	70	70

[1] Specimens conditioned at 23° C. (73.5° F.) and 50 percent relative humidity and tested under these same conditions.

[2] Specimens conditioned in an oven at 50° C. (122° F.) for 48 hours.

[3] Specimens conditioned in an oven at 50° C. (122° F.) for 24 hours.

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at the dividing line of the mold shall be removed.

3.7 Uniformity.—The molded part shall be uniform in color, texture, finish, density, and other physical properties.

3.8 Physical requirements.—Test specimens molded under the conditions specified for the material by the manufacturer of the compound used to mold the molded parts, shall conform to the physical requirements shown in table II.

3.9 Chemical resistance.—The resistance of the molded part to specific chemicals shall be as specified in the contract or purchase order.

3.10 Marking of parts.—Each molded part shall have molded thereon the molder's trademark, the cavity number, and the drawing or part number, unless, because of the small size of the molding, such markings are impracticable. If some but not all of the markings listed can be shown on the molding, the order of preference shall be as listed above.

3.11 Workmanship.—The molded part shall be free from porosity, warpage, checks, cracks, chipped edges, blisters, scratches, and other defects which affect the appearance or which might affect the serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *Molding compound.*—Test specimens shall be molded from the molding compound sampled in such a manner as to obtain a representative sample of a batch of material. A batch of molding compound shall be considered as a unit of manufacture as prepared for shipment and may consist of a blend of two or more production runs of material. The test specimens shall be molded under the conditions specified for the material by the manufacturer of the compound. No special treatment shall be used which will tend to improve the properties of the specimens as compared with those of the molded parts supplied under this specification.

4.1.2 *Molded parts.*—Sampling of molded parts for parts inspection tests shall be as specified in the contract or purchase order.

4.2 Inspection.—The samples of molded parts shall be inspected for type of material, dimensions, inserts, threads, color, finish, uniformity, marking, and workmanship to determine compliance with this specification. Inspection shall also include packaging, packing, and marking for shipment.

4.3 Tests.

4.3.1 *Number of specimens.*

4.3.1.1 *Molding compound.*—One set of test specimens as prescribed in 4.3.6 shall be considered sufficient for testing each batch of molding compound. The average result for the specimens tested shall conform to the requirements prescribed in this specification.

4.3.1.2 *Molded parts.*—The number of specimens for parts inspection shall be as specified in the contract or purchase order.

4.3.2 *Molding compound evaluation.*—Tests for all of the physical requirements listed in table II shall have been made to establish conformity of a material to this specification. These tests may be made by the contractor or by any laboratory designated by the contractor. In the normal course of manufacture these tests will have been made by the material manufacturer. It is not intended that these tests be repeated for each contract or purchase order.

4.3.3 *Molding compound acceptance tests.*—The contractor shall make such tests as specified in the contract or purchase order to determine the acceptability of the compound used to mold the molded parts. Such tests will generally be limited to those required to

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identify the material to the satisfaction of the purchaser. However, the purchaser reserves the right to make, without expense to the contractor, any of the other tests described in this specification to determine that the material meets the requirements of this specification.

4.3.4 *Certification.*—When specified in the contract or purchase order, the contractor shall forward to the procuring agency for each order of molded parts a certified statement (1) identifying the molding compound used by manufacturer's name, trade name, and identification number, (2) certifying that the molding compound used is the same as to manufacturing process and composition as that which was tested in accordance with 4.3.2 and found to conform to all requirements of this specification, (3) presenting the results of the molding compound acceptance tests, and (4) certifying that the molded parts conform to the requirements of the specifications and drawings for the part. The report covering the molding compound evaluation tests shall be referenced on the certified statement.

4.3.5 *Retest and rejection.*—If the results of any test do not conform to the requirements prescribed in this specification, at the option of the purchaser two additional sets of specimens from the same batch or shipment may be tested, each of which shall conform to the requirements specified. If either of these two additional sets of specimens fails, the batch of molded parts may be rejected at the option of the purchaser.

4.3.6 *Test methods.*—The test specimens shall be conditioned and tested at 23° ± 1.1° C. (73.5° ± 2° F.) and 50 ± 4 percent relative humidity in accordance with the requirements of Federal Specification L–P–406, unless other conditions are specified in this specification or in the test method.

4.3.6.1 *Specific gravity.*—Three specimens shall be tested in accordance with method No. 5011 or 5012.

4.3.6.2 *Flexural strength and flexural deflection.*—Five specimens shall be tested in accordance with method No. 1031, using the 5- by 0.5- by 0.25 inch specimen; the 0.25 inch dimension shall be in the direction in which molding pressure is applied and shall be the beam depth of the specimen as tested.

4.3.6.3 *Impact strength, Izod.*—Five specimens shall be tested in accordance with method No. 1071.

4.3.6.4 *Water absorption.*—Three specimens shall be tested in accordance with method No. 7031.

4.3.6.5 *Dielectric strength.*—Five specimens shall be tested in accordance with method No. 4031, except that the test specimens shall be compression molded disks or plates ⅛ inch in thickness. In the short time test, voltage shall be increased at the rate of 1,000 volts per second.

4.3.6.6 *Arc resistance.*—Five specimens shall be tested in accordance with method 4011.

4.3.6.7 *Loss factor.*—Five specimens 4 inches in diameter and ⅛ inch in thickness shall be tested in accordance with method No. 4021.

4.3.6.8 *Chemical resistance.*—Three specimens shall be tested in accordance with method No. 7011.

5. PREPARATION FOR DELIVERY

5.1 **Packaging.**—Unless otherwise specified in the contract or purchase order, the molded parts shall be packaged in accordance with accepted commercial practice. All molded parts shall be individually wrapped or otherwise separated to protect them from abrasion.

5.2 **Packing.**—The molded parts shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest

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applicable rate and to afford maximum protection from normal hazards of transportation. Unless otherwise specified in the contract or purchase order, all molded parts shall be separated by form and size when packed for shipment.

5.3 Marking.

5.3.1 *Package.*—Unless otherwise specified in the contract or purchase order, packages when used shall be marked with the name of the molded part, size, and quantity contained therein, and the names of the contractor and manufacturer.

5.3.2 *Shipping container.*—Unless otherwise specified in the contract or purchase order, each shipping container shall be marked with the following information:

Name of the molded part

Fed. Spec. L–M–181

Type

Form and size (if applicable)

Date of manufacture (month and year)

Color (if other than colorless)

Finish (if other than high-polish)

Quantity, in units as defined in contract or order

Gross weight of container

Name of contractor (and name of manufacturer if not the same)

Government order number (or contract number when order number is not assigned)

Name and address of consignee

Stock No. ---------- (if assigned)

6. NOTES

6.1 Ordering data.—Purchasers should specify required type of melamine molding compound and required design, dimensions, color, opacity, finish, and chemical resistance of the molded part, and should exercise any desired options offered herein. (See 1.2, 3.2, 3.5, 3.6, 3.9, 4.1.2, 4.3.1.2, 4.3.3, 4.3.4, 5.1, 5.2, 5.3, 6.2, and 7.2.4.)

6.2 Bid samples.—It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If samples are required with bid, this requirement should be stated in the inquiry or invitation for bids.

6.3 Transportation description.

6.3.1 *Transportation description applicable to these items is:*

Synthetic resin articles, not otherwise indexed by name, with or without filler.

In barrels or boxes.

Carload weight 24,000 pounds subject to Rule 34, Consolidated Freight Classification.

Plastic articles, not otherwise indexed.

In barrels or boxes.

Truckload weight 20,000 pounds.

Patent notice.—When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and other publications, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

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7.2 **Army, Navy, and Air Force.**

7.2.1 *Applicable specifications and standard.*

Federal Specifications:

NN–B–601—Boxes; Wood-Cleated-Plywood, for Domestic Shipment.

NN–B–621—Boxes; Wood, Nailed and Lock-Corner.

Military Specifications:

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

MIL–A–140—Adhesive, Water-Resistant, Waterproof Barrier-Material.

Navy Department Specification:

General Specifications for Inspection of Material.[1]

Military Standard:

MIL–STD–129—Marking of Shipments.

(Copies of specifications, standards, drawings, and publications, required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Packing.*

7.2.2.1 *Domestic shipment.*—The subject commodity, packaged as described in 5.1, shall be separated by form and size and packed in cleated plywood boxes, or nailed wood boxes, conforming to the requirements of Federal Specifications NN–B–601 and NN–B–621, respectively. The gross weight shall not exceed approximately 200 pounds.

7.2.2.2 *Overseas shipment.*—The subject commodity, packaged as described in 5.1, shall be separated by form and size and packed in cleated plywood boxes, or nailed wood boxes, conforming to the requirements of Military Specifications JAN–P–105 and JAN–P–106, respectively. Shipping containers shall be lined with a sealed waterproof bag, made from material conforming to the requirements of Military Specification JAN–P–125 or the subject commodity shall be wrapped in such material. The seams and closures shall be sealed with a continuous seam at least ¾ inch wide of water-resistant adhesive conforming to the requirements of Military Specification MIL–A–140. The gross weight shall not exceed approximately 200 pounds.

7.2.3 *Marking.*—In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

7.2.4 *Ordering information.*—Procurement documents should specify whether the subject commodity is to be packed and shipping containers marked for domestic or overseas shipment. (See 7.2.2 and 7.2.3.)

7.2.5 *Inspection procedures (Navy only).*—The general inspection procedures shall be in accordance with Navy Department General Specifications for Inspection of Material.

NAVY INTEREST: Md Or S Y

[1] Applicable to Navy purchases only.

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